

Product Description for “Tethys SPC Shape Models and Assessment Products”

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This package contains global and regional topography and albedo data generated by the Gaskell Stereophotoclinometry (SPC) software suite for the Saturnian satellite Tethys. The images used to generate the model are Wide Angle Camera (WAC) and Narrow Angle Camera (NAC) uncalibrated Cassini images from the Cartography and Imaging Sciences Node of NASA's Planetary Data System (PDS) (Porco et al., 2020). Some of the Cassini images have missing or partially missing lines. For Tethys, we replaced these missing pixels with the average of the value of the pixels above and below the missing pixel. This replacement is unlikely to have a substantial effect on the data, but should be noted. A geometric distortion correction is applied to all images. The kernels are from the Navigation and Ancillary Information Facility (NAIF), though a few kernels were “flight kernels” that are no longer available to the public, but should be similar to those archived at NAIF. Images and kernels were added to the model in various stages over the past decade as data came in and the model was improved. The kernels are only used to determine the initial position and pointing of the spacecraft. Afterwards, the position and pointing are updated by SPC as the model improves, so the kernels used are not primary inputs. Images and kernels used to build the global model can be found in document/tethysimagelist.txt and document/tethyskernellist.txt. To aid data users in the quality of the global topography, the tethysshapeassessment.pdf in the document directory contains plots of the assessment data of the global DTM.

Global Model:

Object Name	ICQ Spacing for Q=512 / Q=128	IAU Mean Radius*
Tethys	1.50 / 5.96 km GSD	531.0 km
* From Archinal et al. (2018)		

The native SPC format for a global model is the implicitly connected quadrilateral (ICQ). Each of the quadrilaterals in a model will have the same size, and we refer to this size as “Q”. Larger values of Q will thus give a higher resolution model. The equation to determine the number of vertices from the Q size is $6*(Q+1)^2$ (Gaskell, 2020). We provide the ICQ model with a Q of 512 (1,579,014 vertices), and 128 (99,846 vertices). For each model, the distance between vertices is approximately the same over the whole model, and we refer to this distance as the Ground Sample Distance (GSD). We also provide the global model in other formats, which are discussed below.

Thomas (2010) determined the a, b, and c axes for Tethys. A triaxial ellipsoid generated from the Thomas values match the limbs of Cassini images well, though this SPC model provides a better fit. Nonetheless, there is compatibility between this work and Thomas (2010). A comparison of these two models with an example Cassini image is given in document/tethysvisualcmp.pdf. Note that this type of model to image comparison is routinely performed during SPC processing, and is one of the ways the SPC model is validated.

This global Tethys model is the second archived within the PDS. The first model was generated by R. Gaskell and published in 2013. It included images taken up until 14 Aug 2010 (PDS3 dataset Gaskell, 2013; PDS4 dataset, Gaskell, 2020). Our model is generated using all available images with the last image taken 14 Mar 2017. This model therefore includes ~830

additional images taken after 2010. It also includes an additional ~10 images taken in 2010 or earlier. A full evaluation of this new model to the previous model is outside the scope of this bundle. However, a preliminary examination between these models (Q=512 for both) shows the Root Mean Square (RMS) between the 2013 model and this model is ~1.17 km. Likely the largest contributors to the RMS (which is equivalent to ~0.8 GSD in the Q=512 model), are specific regions of the 2013 model with artifacts that deviate inward from a spherical shape. The largest of these, around 65° Lat and 233° East Lon, is ~170 km wide and has a maximum difference of ~8.2 km. Most of these regions in the 2013 model have smoother topography, though craters are still seen, and likely are the result of poor imagery and/or poor image registration that has since been improved. There is no indication of a center of figure offset between this model and the 2013 model.

Regional Models:

Location Name	Center Lat/East Lon deg	Size (km)	GSD (m)	FormU (m)*
leqs06283	-06 / 283	167.8	335	445.8
lhln19244	19 / 244	268.3	335	485.9
teqs04076	-04 / 076	200.4	400	310.5
thln21112	21 / 112	160.0	235 (300)^	261.0

* See section “Topography and Albedo Data” for details.

^ The effective GSD of this model is 300 m, though it represents topography every 235 m.

The regional models have a higher GSD compared to the global model since they extract more data from the higher resolution images. Regional model image lists can be found in leqs06283imagelist.txt, lhln19244imagelist.txt, teqs04076imagelist.txt, and thln21112imagelist.txt. To aid data users in the quality of the global topography, the tethysregionaldtmassessment.pdf in the document directory contains plots of the assessment data for each regional DTM.

Topography and Albedo Data

The topography and albedo data are provided in various formats, depending upon whether it is global or regional data. For the global model, the topography formats include the native SPC format of Implicitly Connected Quadrilateral (ICQ), plate vector (PLT) and Object Vector (OBJ) formats readable by programs like the Small Body Mapping Tool (SBMT), cubes (.cub) readable by the USGS ISIS program, and geoTiff (.tif) which can be read by GIS programs such as ArcGIS. The global albedo data is provided in cube and geoTiff formats. For the regional models, the formats for topography and albedo are cubes and geoTiff. Note that the topography is the primary product of SPC, while the albedo is a secondary product. The albedo accuracy has not been investigated as thoroughly as the topographic accuracy.

Topographic Accuracy of SPC DTMs:

The topographic error of SPC DTMs was estimated as a part of the OSIRIS-REx mission. As part of NASA Class B software validation, Weirich et al. (2022) used a synthetic, but realistic, digital asteroid model with known heights to produce a simulated mission image suite. Note that this test was performed with an early version of the mission image suite, which had less stereo than the flown mission. With only these images, and no access to the synthetic

asteroid, we generated an SPC global DTM. This global SPC DTM was then compared to the original synthetic DTM to characterize the actual errors. Weirich et al. (2022) showed that a sufficient, but not ideal, image set produced DTMs that were typically accurate to one image pixel of the finest-resolution image. The global imaging campaign for Tethys and the regional DTMs are considered sufficient, so the one image pixel accuracy stated in the previous sentences applies to the products in this bundle. See Weirich et al. (2022) for definitions of a sufficient and ideal image set. Also see Palmer et al. (2022, 2024) and Barnouin et al. (2020) for descriptions of good and poor imagery for SPC. The worst vertex in the test DTM was accurate to about three image pixels of the finest-resolution image. Note that these errors are for both the vertical (i.e. radial) and horizontal directions.

Unlike the OSIRIS-REx testing, we did not always build the Tethys DTMs with a GSD of the finest-resolution image, therefore the accuracy is estimated to be one DTM GSD. Likewise, the worst DTM vertex is estimated to be accurate to three DTM GSD. A good rule of thumb is to estimate the accuracy of the DTMs to be one to two DTM GSD. This gives 1.50 to 3.00 km for the global DTM. For areas where the finest-resolution image is worse than the DTM GSD, a good rule of thumb is to estimate the accuracy to be one to two image pixels, per the previous paragraph. See [document/tethysshapeassessment.pdf](#) and [document/tethysregionaldtmassessment.pdf](#) to see where the image resolution is worse than the DTM GSD. To better understand the accuracy of the topography data, we provide further details in the section titled “Quality Assessment Data” in this document.

SPC has a critical internal consistency check that provides an evaluation of the quality of the DTM, specifically the formal uncertainty, or FormU (Weirich et al., 2022). The FormU indicates how closely the data from the model matches the image data. While the one to two DTM GSD is the estimated accuracy of an individual vertex determined from a test case (i.e. OSIRIS-REx pre-launch testing), FormU is instead a holistic value for a specific DTM. The FormU for the maplets with the best global coverage, that of the 700 m maplets, is 892.0 m. Although FormU is a measure of internal consistency, and not accuracy, Weirich et al. (2022) found the FormU to be within a factor of 2 of the actual error for the OSIRIS-REx testing models. Since the best maplets and the FormU are better than the GSD of the global DTM, we have a good indicator that the error is likely 1 to 2 GSD or 1 to 2 image pixels if the image resolution is worse than the GSD, confirming the numbers in the previous paragraph.

Further complicating matters, the regional DTM thln21112 over-samples the topography in the maplets. For this model the accuracy will be 1 to 2 times the maplet GSD of 300m, or 1 to 2 times the best image resolution for that vertex, whichever is worse.

The FormU for regional models has never been tested like it has been for global models, but we still provide it as an assessment metric. The FormU listed in the Regional Models Table (see table above) is always that of the 300 m maplets. Note that the regional DTM thln21112 over-samples the topography of the maplets. For this model the accuracy will be 1 to 2 times the maplet GSD of 300m, or 1 to 2 times the best image resolution for that vertex, whichever is worse. The FormU is larger than the GSD (or effective GSD in the case of thln21112) for two of the four models. However, most portions of these two models have a GSD that is better than the best image resolution for many of the DTM vertices. This means the image data is oversampled, and the accuracy should be compared to the best image resolution, and not the GSD. Since FormU is a holistic value, the best image resolution also needs to be reduced to a holistic value. To do this we average the best image resolution for each regional model. The average best image resolution is ~496 m for leqs06283 and ~394 m for lhln19244. In this case, the FormU is only larger than the average best image resolution for one model, that of lhln19244. If the FormU for regional models is an accurate predictor of the error of the model (which needs additional testing), then the accuracy of lhln19244 will be slightly worse than the 1 to 2 GSD / “best image pixel” given previously. To estimate the upper bound of the accuracy for lhln19244, we take 2 times the FormU and divide by the average best image resolution, giving a factor of 2.5. This means for lhln19244, the accuracy will be the worse of 1 to 2.5 times the GSD or 1 to 2.5 times the best image resolution. For the other three regional models, the the FormU confirms the rule presented in previous paragraphs.

Formats:

ICQ: The native SPC format for a global model is the implicitly connected quadrilateral (ICQ), and the ICQ models in this bundle have a Q of 512 or 128. The file format is ASCII text with three columns (if topography only) or four columns (if topography and albedo). The ICQ files in this bundle are all four columns. The first three columns are the X (column 1), Y (column 2), and Z (column 3) of the radius vector with units of kilometers, set in the body fixed frame. The +X axis defines 0 degrees longitude, while the +Z axis defines the positive pole (i.e. North Pole for prograde rotation). The fourth column is relative albedo, though we nonetheless still refer to it simply as "albedo". The albedo is a number that varies between 0 (darkest) and 2 (brightest) with the average being 1. Photometric modeling is needed to convert these "albedo" values to an albedo with physical meaning, such as geometric albedo. Further details of the ICQ format and SPC in general, as well as Fortran code to read in the ICQ format, can be found in Gaskell et al. (2008) and Gaskell (2020), respectively. All values are valid, but we still assign a "missing constant" value to the xml label.

All ICQ files end with an "i" to distinguish them from other files.

PLT and OBJ: As previously stated, PLT is a plate model, while OBJ is an object vector readable by the SBMT. These files were generated using the Applied Physics Lab Altimetry Working Group tools generated for the OSIRIS-REx mission. The programs used are ICQ2PLT and PLT2OBJ (see Acknowledgements section). The OBJ files generated by PLT2OBJ have been modified to reduce the number of significant figures and the columns have been padded with white spaces to align all columns. Both formats are in units of kilometers. All values are valid, but we still assign a "missing constant" value to the xml label.

All PLT files end with a "p" to distinguish them from other files.

All OBJ files end with a "o" to distinguish them from other files.

GeoTiff: The topography and albedo of the Q=512 model are provided as separate geoTiffs both with equirectangular and polar stereographic projections and with square pixels. Three geoTiffs are provided for the topography; an equirectangular projection from -55° to 55° latitude, a north polar stereographic projection from 90° to 55° latitude, and a south polar stereographic projection from -90° to -55° latitude. Three geoTiffs for the albedo are similarly provided. The SPC ICQ data was converted to geoTiffs using Geospatial Data Abstraction Library (GDAL) tools. The GDAL program ogr2ogr was used to project the data and gdal_grid, with an inverse distance interpolation algorithm, was used to convert the projected data into a geoTiff. The geoTiffs are resampled values from the ICQ generated by SPC, and are in units of meters. The horizontal spacing is kept close to that of the ICQ GSD. These files are 32-bit, which is required to obtain the spatial resolutions needed for some solar system bodies. To provide data product similarity across all bodies, we always generate 32-bit geoTiffs even when high spatial resolution is not needed. 32-bit geoTiff files cannot be read by simple programs and require programs such as ArcGIS to be accessed. We assign a "missing constant" value to the xml label.

The regional models are also provided as geoTiffs, generated with the same procedures as the equatorial geoTiffs. However instead of the topography cube representing radius, it is instead the deviation from Tethys' datum of 531 km.

All geoTiff files end with a "g" to distinguish them from other files.

Cubes: The topography and albedo of the Q=512 model are provided as separate cubes, both with equirectangular and polar stereographic projections and square pixels. Three cubes are provided for the topography; an equirectangular projection from -55° to 55° latitude, a north polar stereographic projection from 90° to 55° latitude, and a south polar stereographic projection from -90° to -55° latitude. Three cubes for the albedo are similarly provided. The cubes were generated from the geoTiffs using the GDAL program gdal_translate. As such, they

are also resampled values from the ICQ generated by SPC. The equatorial cube geometry block was set using ISIS maptemplate and maplab. The polar cubes geometry block was set using GDAL. The topography is given as radius in meters. The horizontal spacing is kept close to that of the ICQ GSD. We assign a “missing constant” value to the xml label.

The topography and albedo of the regional models are generated using a different pipeline from that of the global cubes. The cubes are not projected but approximate north up. Orientation vectors are given in the PDS label under local_georeference_information. The cubes are given in "SPC space" so that resampling of the SPC data does not occur. The horizontal spacing between the vertices is the GSD of the original DTM. Three cubes fully define the topography. These are topography, latitude, and longitude cubes. The topography cube is the elevation value in meters from Tethys' datum of 531 km, the latitude cube is the Latitude value of that vertex in degrees, and the longitude cube is the East Longitude value of that vertex in degrees. A fourth cube is for the relative albedo of each vertex, so the latitude and longitude cubes apply to albedo as well as topography. Note that all values in the topography, latitude, longitude, and albedo cube files are valid, but we still assign a “missing constant” value to the xml label.

All Cube files end with a "c" to distinguish them from other files.

Global DTM Quality Assessment Data

We provide metadata for the global DTM to understand the quality of the topography and albedo data, as well as to provide information about where future high-resolution regional DTMs could be generated and the expected quality of their data. This metadata is in the form of sigma values of the topography, best maplet resolution, number of images, and best image resolution. All quality assessment data is provided in USGS ISIS cube format. We provide images of these products, along with our assessment of each, in document/tethysshapeassessment.pdf.

To better understand the sigma values and best maplet resolution, a brief description of an SPC maplet is given here. Further details can be found in Palmer et al. (2022). A maplet is a DTM that is usually 99 x 99 vertices and represents a portion of the surface. Maplets can have any vertex spacing, also called ground sample distance (GSD), and are placed to overlap with other maplets. Initial stages of the global DTM are made with large maplets, and as the global DTM develops, smaller and smaller maplets are used. The result is every portion of the surface has multiple maplets with many different GSD values.

Similar to the global radius and albedo in cube format, all assessment data uses an equirectangular projection with square pixels. All assessment products are in 1-degree bins, and have 360 by 180 pixels.

Best Maplet Resolution:

The best maplet resolution is given in units of meters in cube format. Note that the maplet resolution for SPC models can be any value, and this value can be better or worse than the GSD of the global DTM, and better or worse than the best image resolution. In the case of this Tethys model the maplet GSD is always better (i.e. smaller) than the global DTM. A “missing constant” value is assigned in the xml label because there are some sampling gaps near the poles.

Sigma Values:

The sigma value is a measure of how well all the maplets agree at a particular vertex on the global DTM, with smaller values indicating better agreement. Note that the sigma is not a measure of the error of the radius at that vertex, rather it is a measure of internal consistency

between the model and the images. Nonetheless, the more the maplets differ in their height, the less likely the final radius will be close to the truth.

SPC returns the sigma value in meters for each vertex of the global DTM. We downsample these sigma values into 1-degree bins using the same GDAL process as the radius and albedo files. Because of the projection, values near the poles often show missing values, nonetheless the data constitutes complete coverage of the object. We assign a “missing constant” value in the xml label.

Best Image Resolution:

Best image resolution refers to the best or finest-resolution image used at every vertex. The best image resolution is given as meters. All values are valid, but we assign a “missing constant” value in the xml label.

Number of Images:

Typically more images indicate more accurate heights. SPC has a theoretical lower limit of three images, though in practice five images are an effective lower limit due to less than optimum observation geometries. Areas with fewer than five images should be treated with caution. At the scale of this Tethys model, all areas have five or more images, indicating accurate heights. In general, the number of images is the least useful quality assessment since it does not capture the uniqueness of the viewing conditions. An example of a poor observation set is one in which there are five images taken one second apart. The sun vector has not changed and the spacecraft vector has not changed appreciably, so those five images are effectively one unique image. Another example is a single spacecraft flyby with ten images of the same region on the surface, versus ten spacecraft flybys from different angles, each with a single image of the same surface. The latter is far better than the former even though they each have ten images. The latter is better because it will have a larger variety of emission and incidence angles (see Palmer et al. 2024 and Barnouin et al. 2020 for details, and a description of good and poor imagery). Another limitation to be aware of is that low resolution images are counted even if they do not contribute to the solution. An example is five good images at 300 m/px, and fifty images at 3,000 m/px. The number of images will be 55, even though effectively there are only five images contributing to the solution.

Nonetheless, we feel it is still a good qualitative measure of quality, and can highlight low quality areas of the surface that would otherwise appear to be high quality. An example is a portion of the surface with high image resolution, but only three images.

All values are valid, but we assign a “missing constant” value in the xml label.

Formats

Cubes: All global assessment data is provided in USGS ISIS cubes, as well as thumbnails in document/tethysshapeassessment.pdf. As stated above, all global assessment products are given in 1-degree bins with an equirectangular projection. All cube files end with a "c" to distinguish them from other files.

Regional DTM Quality Assessment Data

Like the global assessment data, the regional DTM quality assessment data is provided in the form of sigma values of the topography, number of images, and best image resolution. The best maplet resolution is not provided because all maplets in the regional DTMs have a 300 m GSD. The assessment data is provided in cube form like the albedo and topography cubes. Also like the albedo and topography cubes, the data is not projected, but the latitude and longitude cubes define the location of each value in the assessment data. We provide images of these products, along with our assessment of each regional model, in document/tethysregionaldtmassessment.pdf.

All values are valid, but we assign a “missing constant” value in the xml label.

Formats

Cubes: All regional assessment data is provided in USGS ISIS cubes, as well as thumbnails in document/tethysregionaldtmassessment.pdf. All cube files end with a "c" to distinguish them from other files.

Photometric Data

We generated photometric data (in cube format) that is aligned to and has the same horizontal spacing as the topographic cube data. Fewer images were used for photometric data than for topographic data. For each Cassini image used for photometric data, four cubes were generated; I/F (reflectance), Local Phase Angle, Local Emission Angle, and Local Incidence Angle. Part of the SPC process is to align each image to the topography by adjusting the position and pointing of the spacecraft. The alignment is typically accurate to the size of 1 GSD or better, though rarely for some images in localized areas the misalignment may be as large as 3 or 4 GSD. All angles were determined by the Sun and spacecraft position relative to the height of each DTM vertex. All cubes of a particular regional DTM are in the same physical space. In other words, sample, line (100,100) of the topography cube maps to sample, line (100,100) in all the other cubes giving I/F, local phase, incidence, and emission angles for all images. This is not only true of the photometric data and the topographic data, but also of the assessment data.

Note that while some cubes have values for all pixels, only pixels that correspond to non-zero values in the reflectance cube (i.e. I/F or “ioverf”) are valid. Sample/line values that are zero in the reflectance cube are also invalid for the phase, emission, and incidence angles.

Formats

Cubes: All photometric data is provided in USGS ISIS cubes. All cube files end with a "c" to distinguish them from other files.

File Naming Schema

All global product filenames begin with “tethys”, followed by a <type> identifier. For the table files, the <type> identifier is “128” for the Q=128 model or “512” for the Q=512 model. For the cube and geoTiff global model the <type> identifier is “eq<prod>” for the equatorial product, “np<prod>” for the north polar product, or “sp<prod>” for the south polar product, where <prod> is either “radius” or “albedo”. For the remaining global products the <type> identifier is “sigma”, “bestmap” for the best maplet resolution, “bestimg” for the best image resolution, and “numimg” for the number of images. The <format> identifier is one character to make the xml labels unique. The <format> identifiers for the table files are “i”, “p”, and “o”, for

ICQ, plate vector, and object vector, respectively. For remaining global products "c" will be used for cubes, "g" will be used for geoTiff.

Global examples:

tethys_512_i.tab

tethys_bestmap_c.cub

All regional product filenames in this bundle begin with a <location> identifier. This identifier has both region and central lat/lon DTM information. The first three characters will be of an abbreviated version of the region, the next three for the latitude, and the next three for longitude. There are four regions and the abbreviations are "leq" for leading equator, "lhl" for leading high latitude, "teq" for trailing equator, and "thl" for trailing high latitude. The latitude is represented by an "n" for North and an "s" for South, followed by the numeric value of the latitude accurate to a whole number. The longitude is represented by the numeric value in East Longitude accurate to a whole number.

Location examples:

Leading Equator centered at -6 Lat, 283 E Lon becomes leqs06283

Trailing High Latitude centered at 21 Lat, 112 E Lon becomes thln21112

For photometric data, the next identifier is the Cassini image name, which is 11 characters.

Example: N1558909397

The next identifier is <type>, which will be "topo" for topography, "lat" for latitude, "lon" for longitude, "alb" for albedo, "ioverf" for I/F, "a" for phase, "e" for emission, "i" for incidence, "bestimg" for best image resolution, "numimg" for number of images, and "sigma" for the sigma.

The last identifier is one character to make the cube and geoTiff xml labels unique. "c" will be used for cubes, while "g" will be for geoTiff.

Thus, all topography, latitude, longitude, albedo, best image, number of images, and sigma cubes will have the filename structure: <location>_<type>_c.cub

Regional Examples:

lhln19244_topo_g.tif

leqs06283_numimg_c.cub

leqs06283_N1558909397_a_c.cub

Locations

This bundle contains 4 regional DTMs. These locations were chosen to provide example surfaces to study the variations in photometric properties as a function of radiation environment. Tethys is tidally locked to Saturn, the leading half of Tethys is always moving into the direction of motion, and thus sweeps ions from the magnetic field onto its surface. The trailing hemisphere trails the direction of motion, and is thus more shielded by the satellite from the ions traveling along Saturn's magnetic field lines. Radiation variations also exist from equator to pole. Thus, four regions, with similar morphological characteristics were selected: two from the leading hemisphere, one high-latitude and one equatorial, and two from the trailing hemisphere, likewise one high-latitude and one equatorial region. The regions were selected to best match morphology and optimize image coverage.

Acknowledgments

The programs used to generate the PLT and OBJ files were from the altimetry working group (AltWG) of the OSIRIS-REx mission. While these tools are not currently available to the public, a newer version of the programs, ShapeFormatConverter, is available on GitHub as part of the “Terrasaur” software package (<https://github.com/JHUAPL/Terrasaur>).

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